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Study Notes

Insect Toxicology

Introduction

Definition of Toxicology

"Toxicology is the branch of medical science that deals with the nature, properties, effects and the detection of poison. It is, therefore, the science of poisons" (Du Bois and Geiling, 1959).

"The study of limits of the biological effects of a chemical or mixtures of chemicals" (Fogleman, 1963).

Factors Influencing Toxicity

- **Nature of Toxicant:** The type of toxicant used significantly influences its toxicity.
- **Routes of Exposure:** Toxicity can vary based on how the toxicant enters the organism—oral, dermal, or inhalation.
- **Dose:** Toxicity is dose-dependent, and commonly expressed as LD50 or LC50 values.
- **Organism:** Different species may react differently to the same toxicant.

Toxicity Parameters

There are several ways to quantify a chemical's toxicity to an organism, some of which are described here.

1. LD_{50} (Median Lethal Dose)

- LD_{50} is the dose that kills 50% of the test population under specific experimental conditions.
- It is typically expressed as milligrams of poison per kilogram of body weight and can vary based on the route of exposure (oral, dermal, inhalation).
- **Smaller LD_{50} values indicate higher toxicity**.

Categories based on LD50 values:

Category	LD_{50} Range (mg/kg)	Examples
Extremely Toxic	1 mg/kg or less	Parathion, Aldicarb
Highly Toxic	1-50 mg/kg	Endrin
Moderately Toxic	50-500 mg/kg	Carbofuran
Slightly Toxic	500-1000 mg/kg	Malathion
Non-toxic (Practically)	1-5 gm/kg	

2. LC_{50} (Median Lethal Concentration)

- LC_{50} is the concentration causing 50% mortality in the external medium (air or water).
- Used when exact individual doses are challenging to determine.
- **Expressed as a percentage of active ingredient or parts per million (ppm).**

3. LT_{50} (Median Lethal Time):

- LT_{50} represents the time needed to cause the death of 50% of a population at a specific dose or concentration.
- Expressed in hours or minutes.

- Widely used in field studies and testing insect viruses, particularly NPV (Nuclear Polyhedrosis Virus).

4. KD_{50} (Median Knockdown Dose)

- Represents the dose of an insecticide or the time required to knock down 50% of the insect population

5. KT_{50} (Median Knockdown Time):

- Signifies the median time needed to achieve a 50% knockdown of insects.
- Primarily employed in the evaluation of synthetic pyrethroids against insect populations.

6. ED_{50} (Median Effective Dose)

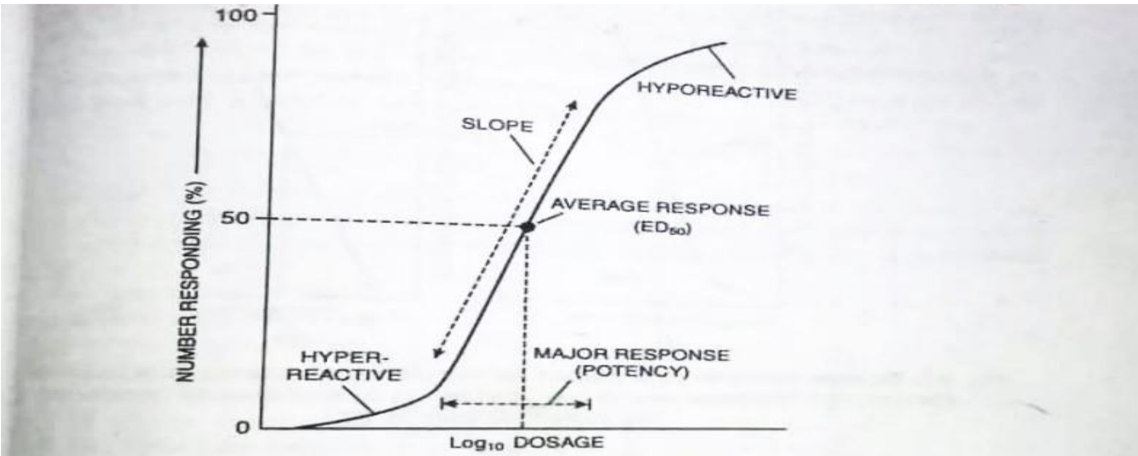
- Refers to the median effective dose, indicating the dose of a chemical required to produce a desired effect in 50% of the population

7. EC_{50} (Median Effective Concentration):

- Represents the median effective concentration, denoting the concentration of a chemical needed to achieve a specific effect in 50% of the population.
  Both are mainly utilized in assessing the effectiveness of insect growth regulators (IGR).

Dose-Response Relationships:

- Toxicity is determined by quantifying the response to a series of increasing doses.
- To study the toxic effects of any substance, **a graph is drawn in which number of individuals responding are plotted against the dosage (Log10 dosage)**, thus a sigmoid is obtained which is known as **dosage response curve**.
- Assumptions include a threshold dose, increasing response with dose, and a maximum response point.



9.4. Typical sigmoid cumulative dosage response curve for a toxic effect, which ymmetrical about the average (50 per cent response) point. The major response otency) occurs around the average response. The slope of the curve is determined by the increase in response as a function of incremental increases in dosage. Hyper-reactive and hypo-reactive individuals are found at the extreme left and right sides of the curve, respectively

Methods of Testing Chemicals on Insects

- **Topical Application**: Applied directly to the body surface, expressed in µg AI/insect.

- **Injection Method:** Used when precise amounts inside the insect are necessary.

- **Dipping Method:** Practical for small insects or eggs, with results expressed as LC_{50}.

- **Contact or Residual Method:** Involves applying the insecticide to a surface and allowing it to evaporate, expressing results as mg or g AI/m2.

- **Feeding and Drinking Method:** Evaluates toxicity of ingested chemicals, with unlimited or limited availability.

Evaluation of Toxicity in Insects and Animals

- **Probabilistic Approach**: Probit units of percent mortalities plotted against a logarithmic scale of dosages. This method results a straight line which facilitates the determination of the LD 50 and other values on the plot.

- **Adjusted Mortality**: To the natural mortality in controls and to get more accurate results, Abbott's formula (Abbott,1925) is used,

$$\boxed{\textbf{Corrected mortality percent} = \frac{(P-Po)}{(100-Po)} \times 100}$$

Where, **P= Percent Mortality**, **P_o =Percent mortality of insect in the untreated control**, This adjusted value is permissible when mortality in the controls does not exceed 20%.

- **Toxicology in Higher Animals:** Emphasis on qualitative aspects, with LD_{50} values often determined in rats or mice.

Insecticides

- Insecticides are chemical substances designed to control insect populations by either killing them or disrupting their harmful activities.
- They play a crucial role in safeguarding crops, plants, and public health from insect-relateddamages and diseases.
- Insecticides are classified based on their chemical structure and the way they act on insects. These modes of action determine how the insecticides affect the insects and achieve control.

Applications:

- Insecticides are widely used in agriculture to protect crops from insect pests, preventingyield losses and ensuring food security.
- Insecticides are employed to control disease-carrying vectors like mosquitoes, reducingthe spread of diseases like malaria and dengue.
- Insecticides find applications in industries, controlling pests that can damage stored products

- or affect manufacturing processes.
- These are used to manage pests like roaches and termites in homes, businesses, andpublic spaces.


Classification of pesticidesClassification of pesticides based on target organisms

a) Insecticides eg. endosulfan, malathion
b) Rodenticides eg. Zinc phosphide, warfarin
c) Acaricides eg. dicofol, azinphos methyl
d) Avicides eg. TMTD, anthraquinone
e) Molluscides eg. metaldehyde, trifenmorph
f) Nematicides eg. DD, ethylene dibromide
g) Fungicides eg. Copper oxychloride, mancozeb
h) Bactericides eg. Streptomycin sulphate, aureomycin
i) Herbicides eg. 2,4-D, butachlor

Classification Based on Chemical Nature:

A. **Inorganic Insecticides:** Examples include compounds containing arsenic and fluorine. These substances are used for insect control.

B. **Organic Insecticides:** These are categorized further:
   a) Hydrocarbon Oils: Oils used to control pests through physical mechanisms.
   b) Animal-Origin Insecticides: Derived from animals, such as neristoxin.
   c) Plant-Origin Insecticides: Derived from plants, like nicotine, pyrethrum, rotenone, and neem.
   d) Synthetic Organic Compounds: Human-made compounds:

Classification Based on Mode of Entry:

A. **Stomach Poison:** Insects ingest toxins through food (e.g., B.t).
B. **Contact Poison:** Insects die upon physical contact (e.g., chlorinated hydrocarbons).

C. **Fumigant:** Chemicals used to create a gaseous form to control pests (e.g., DDVP, Lindane).
D. **Systemic Poison:** Insecticides are absorbed by plants and affect insects feeding on treated plants (e.g., methyl demeton, dimethoate).

Classification Based on Mode of Action:

A. **Physical Poison:** Inert dusts with physical impact.
B. **Protoplasmic Poison:** Heavy metals (mercury, copper), arsenic, and fluorine thatdisrupt protoplasm.
C. **Respiratory Poison:** Chemicals like hydrogen cyanide and carbon monoxide affecting respiration.
D. **Nerve Poison:** Affects nervous systems (organophosphates, carbamates).

Insecticide formulations

- Formulation involves processing of the technical grade insecticides for better storage, handling, measure, application and efficacy together with safety
- Depending upon the mode of applications, dry and liquid formulations are common forms
- They may also be classified as solid, liquid and gaseous formulations.

EC: Emulsifiable concentrate

DC: Dispersible concentrate

G –Granule

TB - Tablet

SB-Soluble concentrate

GL- Emulsifiable gel

OL-Oil miscible liquid

WSC –Water soluble concentrate

Classification	Formulations
Solid Formulations	Dust, Wettable or Water Dispersible Powder, Granules, Capsules and Baits
Liquid Formulations	Emulsifiable Concentrate, Ultra Low Volume Formulations, Suspension
Gaseous Formulations	Fumigants, Aerosol, Foams, Smokes, Mists and Fog

Characteristics and Functions of Insecticides

Organochlorine (OC) Insecticides:

Organochlorine insecticides belong to a class of pesticides characterized by the presence of carbon, chlorine, and hydrogen atoms. They can be classified into various groups based on their chemical structures:

Cyclodiene Series:
- **Examples:** Chlordane, Aldrin, Heptachlor, Endosulfan, Dieldrin, Isodrin, Endrin.
- Very stable in the environment, forming epoxides that contribute to their high persistence.

Halogenated Aromatic Compounds:
- **Examples:** DDT, DDE, Dicofol, Methoxychlor.
- DDT is a well-known organochlorine insecticide; DDE and Methoxychlor are derivatives.

Cycloparaffins:
- **Example:** BHC/HCH (Benzene Hexachloride, Lindane).
- BHC includes alpha, beta, gamma, delta, and epsilon isomers; Lindane is the gamma-isomer.

Chlorinated Terpenes:
Example: Polychlorpinenes.

Specific Organochlorines:

1) DDT (Dichlorodiphenyltrichloroethane):
- Structure: 2,2-bis-(p-chlorodipheny)-1,1,1-trichloroethane.
- Appearance: White crystalline solid.
- Solubility: Soluble in various solvents.
- Toxicity: Central nervous system stimulant, causing hyperactivity and tremors.
- Environmental Impact: Highly persistent, accumulates in body fat, and has a long-lasting effect.

2) BHC/HCH (Benzene Hexachloride, Lindane):
- Structure: 1,2,3,4,5,6-hexachlorocyclohexane.
- Appearance: Off-white to brown, amorphous powder.
- Toxicity: Isomer-dependent, with alpha and gamma isomers acting as central nervous system stimulants.
- Environmental Impact: Highly persistent in soils, stable in water environments.

3) Lindane:
- Structure: 1α,2α,3β,4α,5α,6β-hexachlorocyclohexane.
- Appearance: Colorless crystals.
- Toxicity: CNS stimulant, causing convulsions.
- Environmental Impact: Persistent in soils, shorter half-life when sprayed on the surface.

4) Endosulfan:

- Structure: 6,7,8,9,10,10-hexachloro-1,5,5a,6,9,9a-hexahydro-6,9-methano-2,4,3-benzadioxathiepin 3-oxide.
- Appearance: Yellow-brown color.
- Toxicity: Mixture of alpha and beta isomers, acting as contact poison for various insects and mites.
- **Environmental Impact: Moderately persistent in soil, breakdown by fungi and bacteria.**

Organophosphorous Insecticides:

- Neutral ester or amide derivatives of phosphorous acids with phosphoryl (P-O) or thiophosphoryl (P-S) groups.
- Manufactured at high temperatures, containing isomers or by-products with anticholinesterase activity.
- Non-persistent and biodegradable.

Pyrophosphates:

- **Examples: Schradan, TEPP.**
- Other Groups: Dialkylarylphosphate, Phosphorothioate, Phosphorothionate, Phosphorothiolate, Phosphorothiolothioate.
- Majority of organophosphorus pesticides belong to this group, highly toxic with LD_{50} values ranging from 5-55 mg/kg body weight. Metabolites can be more toxic than the parent compounds.

Phosphorohalides and Cyanides:

- Examples: Nerve gases such as Mipafox.

Specific Organophosphorus Pesticides:

1) Malathion (Diethyl (dimethoxy thiophosphorylthio) succinate):

- Nonsystemic, wide-spectrum organophosphate insecticide.
- Appearance: Clear, amber liquid.
- Solubility: Miscible with most organic solvents.
- Toxicity: Relatively low toxicity with an LD_{50} of 2800 mg/kg.
- **Environmental Impact: Low persistence in soil, rapid degradation related to soil binding.**

2) Parathion (O,O-diethyl O-4-nitrophenyl phosphorothioate):

- Broad-spectrum organophosphate pesticide.
- Produced by condensation of O,O-diethyl phosphorochloridothioate with sodium 4-nitrophenoxide.
- Appearance: Pale yellow liquid (pure), deep brown to yellow liquid (technical).

- Solubility: Slightly soluble in petroleum oils, miscible with most organic solvents.
- Toxicity: Relatively high toxicity with low LD_{50} values (e.g., 13 mg/kg in male rats).
- **Environmental Impact: Limited potential for groundwater contamination, binds tightly to soil particles, degraded by biological and chemical processes within weeks.**
- **Photodegradation and conversion to more toxic metabolites (e.g., paraoxon) can occur.**

Carbamates:

- Synthetic derivatives of physostigmine, esters of carbamic acid.
- Low toxicity to mammals, broad spectrum for insect control.
- Functions:
- Act as anticholinesterase inhibitors, causing twitching, incoordination, convulsions, and paralysis.
- Examples: Carbaryl (Sevin), Aldicarb (Temik).

Carbaryl (1-naphthyl methylcarbamate):

- Carbaryl is a contact insecticide belonging to the carbamate group with slight systemic properties.
- Produced by the reaction of 1-naphthol with methyl isocyanate or with carbonyl chloride and methylamine.
- **Trade Names:** Carbaryl, Adios, Bugmaster, Carbamec, Carbamine, Crunch, Denapon, Dicarbam.
- Solid, color varies from colorless to white or gray, with pure carbaryl being a colorless crystalline solid.
- Empirical Formula: $C_{12}H_{11}NO_2$.
- **Common Formulations:** Emulsifiable concentrate, granules, wettable and dustable powder, bait pellets, micronized suspensions in molasses, in non-phytotoxic oil, or in aqueous media.
- Breakdown in Soil and Groundwater:
- **Persistence in Soil: Carbaryl has low persistence in soil**.
- Degradation: Degradation in soil is primarily due to sunlight and bacterial action.
- Environmental Impact: Sunlight initiates the breakdown of carbaryl in the soil, and bacterial action further contributes to its degradation.

Pyrethroids:

- Natural esters from chrysanthemum flowers or synthetic pyrethroids introduced in the 1970s.
- More stable, higher insecticidal activity, and lipophilic in nature.
- Nerve poisons, affect nerve axon, causing repetitive discharge, paralysis, and death.
- Examples: Allethrin, Permethrin, Cypermethrin.

Cypermethrin:

- Cypermethrin is a synthetic pyrethroid known for its stomach and contact action.
- Produced by the esterification of alpha-hydroxy-3-phenoxy-phenylacetonitrile with 3-(2,2-dichlorovinyl)-2,2-dimethylcyclopropanecarboxylic acid.

- Trade Names: Ammo, Arrivo, Barricade, Basathrin, and Super.
- Appearance: Pure isomers form colorless crystals; mixed isomers result in a viscous semi-solid or a viscous, yellow liquid.
- Empirical Formula: $C_{22}H_{19}Cl_2NO_3$.
- A mixture of eight different isomers, each with potentially different chemical and biological properties.
- Available as an emulsifiable concentrate or wettable powder.
- Breakdown in Soil and Groundwater:
- **Persistence in Soils: Moderately persistent; in aerobic conditions, the soil half-life ranges from 4 days to 8 weeks.**
- Laboratory Conditions: Applied to sandy soil, laboratory conditions showed a half-life of 2.5 weeks.
- **Environmental Impact:** Cypermethrin undergoes degradation in soils, contributing to its moderate persistence.

Botanical Insecticides:

- **Derived from plants, utilizing secondary metabolites.**
- Safe to natural enemies, biodegradable, and leave minimal toxic residues.
- **Nicotine:** Acts as a nerve poison, mimics acetylcholine.
- **Rotenone**: Inhibits respiratory metabolism, used against external parasites of livestock.
- Others: Neem, Chinaberry, Pongam, Custard apple, Ryania, Limonene, Sabadilla.

Bioinsecticides:

- Use of natural biological control agents, including **parasites, parasitoids, predators, and pathogens**.
- Introduction, conservation, and augmentation methods.
- Utilizes microbial control with specific targeting.
- **Used for Inoculative releases and inundative releases.**

Neonicotinoids:

- Synthetic **derivatives from nicotinoids, systemic poisons**.
- Act as **agonists at insect nicotinic acetylcholine receptors (nAChR).**
- Higher toxicity to insects compared to nicotinoids.
- Examples: Imidacloprid, Acetamiprid.

Fumigants:

- Gaseous insecticides, slightly heavier than air, spread to all areas of a sealed structure.
- Ideally easily generated, harmless to foods, non-explosive, and non-persistent.
- Mode of action varies, may kill rapidly or slowly.
- Examples: Ethylene dibromide.

Insecticide Types and Their Mode of Action

Insecticide Type	Mode of Action
Organochlorine	Act on neurons, causing sodium/potassium imbalance for nerve impulse transmission. Some affect GABA receptors, causing hyperexcitable state, convulsions. Broad-spectrum, no longer widely used.
Organophosphate	Cause acetylcholinesterase (AChE) inhibition and accumulation of acetylcholine at neuromuscular junctions causing rapid twitching of voluntary muscles and eventually paralysis. A broad-range insecticide, generally the most toxic of all pesticides to vertebrates.
Organosulfur	Exhibit ovicidal activity (i.e., they kill the egg stage). Used only against mites with very low toxicity to other organisms.
Carbamates	Cause acetylcholinesterase (AChE) inhibition leading to central nervous system effects (i.e., rapid twitching of voluntary muscles and eventually paralysis). Has very broad spectrum toxicity and is highly toxic to fish.
Formamidines	Inhibit the enzyme monoamine oxidase that degrades neurotransmitters causing an accumulation of these compounds; affected insects become quiescent and die. Used in the control of OP and carbamate-resistant pests.
Dinitrophenols	Act by uncoupling or inhibiting oxidative phosphorylation preventing the formation of adenosine triphosphate (ATP). All types have been withdrawn from use.
Organotins	Inhibit phosphorylation at the site of dinitrophenol uncoupling,preventing the formation of ATP. Used extensively against mites on fruit trees and formerly used as an antifouling agent and molluscacide; very toxic to aquatic life.
Pyrethroids	Acts by keeping open the sodium channels in neuronal membranes affecting both the peripheral and central nervous systems causing a hyper-excitable state. Symptoms include tremors, incoordination, hyperactivity and paralysis.

Nicotinoids	Act on the central nervous system causing irreversible blockageof the postsynaptic nicotinergic acetylcholine receptors.
Spinosyns	Acts by disrupting binding of acetylcholine in nicotinicacetylcholine receptors at the postsynaptic cell.
Pyrazoles	Inhibit mitochondrial electron transport, ATP disruption.Effective against psylla, aphids, whitefly, thrips.
Pyridazinones	Interrupt mitochondrial electron transport at Site 1. Mainly usedas miticide. Toxic to aquatic arthropods and fish.
Quinazolines	Act on larval stages by inhibiting chitin synthesis. Rupture,starvation. Not registered in U.S.
Botanicals	Effects vary: pyrethrum stimulates nerve cells, nicotine mimics Ach, rotenone inhibits respiration, limonene affects sensorynerves, neem disrupts molting.
Synergists/Activators	Inhibit P-450 enzymes, enhance insecticide activity. Not toxicthemselves.
Antibiotics	Block GABA, stop feeding, egg laying; control mites, leafminers.
Fumigants	Act as narcotics, induce narcosis or unconsciousness. Effectvaries.
Inorganics	Mode varies: uncoupling, enzyme inhibition, desiccation. Impactvaries.
Biorational	Biochemicals (hormones, enzymes) or microbials (viruses, bacteria). Attractants, regulators, endotoxins. Low toxicity.
Benzoylureas	Act as insect growth regulators by interfering with chitin synthesis.

Insecticides and their Trade Name:

Chemical	Trade Name
O-S Dimethyl acetylphosphoramidothioate	Acephate/Asatof/Starthene
Thiamethoxam	Actara

Imidacloprid	Admire, Confidor
Indoxacarb	Avaunt
Chlorantraniliprole	Corogen
Cypermethrin	Cymbush
Deltamethrin	Decis
Chlorpyrifos	Dursban
Quinolphos	Ekalux
Flubendiamide	Fame
Carbofuron	Furadon/Tata
Triazophos	Hostathian
Fipronil	Jump
Dicofol	Colonel-S
Methomyl	Lannate
Melathion	Malathion
Dimethoate	Rogor
Carbaryl	Sevin
Fipronil	Regent
Acetamiprid	Pride/Rekord
Phorate	Thimet

Toxicological Symptoms of Pesticide Exposure

Introduction

- Pesticides, a diverse group of substances, can enter the human body through absorption, ingestion, or inhalation.
- The toxic effects of pesticides vary, and exposure can occur dermally, orally, or through inhalation.

1. **Dermal Exposure:**
- Absorption occurs immediately upon contact with the skin or eyes.
- The rate of dermal absorption varies for different body parts.
- Relative absorption rates are determined by comparing with the forearm absorption rate.

2. **Oral Exposure:**
- May result in serious illness, severe injury, or death if pesticides are swallowed accidentally, carelessly, or intentionally.

3. **Respiratory Exposure:**
- Hazardous as pesticide molecules can be rapidly absorbed by the lungs into the bloodstream.
- Can cause serious damage to nose, throat, and lung tissue.
- Risks are higher with vapors and small pesticide amounts.
- High-pressure, ultra-low volume, or fogging equipment increases the potential for respiratory exposure.

Toxicity Terms for Mammals

- **Acute Toxicity**: Immediate, severe effects from a single high-dose exposure.
  Example: Rapid illness or fatality due to concentrated toxic substance exposure.

- **Chronic Toxicity**: Adverse effects from prolonged exposure to low levels of a toxicant.
  Example: Long-term health issues from sustained, low-level pesticide exposure.

- **Oral Toxicity:** Effects from consuming a pesticide orally.
  Example: Health problems due to ingesting a pesticide-contaminated substance.

- **Dermal Toxicity:** Effects from skin absorption of an insecticide.
  Example: Irritation or systemic effects from skin contact with pesticides.

- **Inhalation Toxicity:** Effects from inhaling fumes or vapors, often with fumigants.
  Example: Respiratory issues from breathing in fumes of a toxic substance.

Signs and Symptoms

- **Poisoning signs (observable):** Vomiting, sweating, pin-point pupils.
- **Symptoms (functional changes):** Nausea, headache, weakness, dizziness.

Poisoning of Organochlorine Pesticides:
- Not readily biodegradable, affecting the nervous system as stimulants or convulsants.
- Symptoms include apprehension, excitability, dizziness, headache, muscle twitching, loss of coordination, convulsions, unconsciousness.
- No specific antidotes; decontamination involves removing contaminated clothing, washing with soap and water.

Poisoning of Organophosphorus Pesticides:

- Inhibit acetylcholinesterase, affecting the nervous system.
- Symptoms include headache, fatigue, nausea, blurred vision, slowed heartbeat.
- Mild, moderate, and severe exposures have distinct symptoms.
- Antidotes (e.g., atropine, pralidoxime) available for emergency treatment.

Poisoning of Carbamate Pesticides:

- Cholinesterase inhibitors with symptoms ranging from constricted pupils to diarrhea and stomach pain.
-

Poisoning of Synthetic Pyrethroids:

- Generally low toxicity, with symptoms like incoordination, tremors, salivation, vomiting at very large doses.
- Skin irritation and asthma can occur; systemic toxicity by inhalation and dermal absorption is low.

Poisoning of Bioinsecticides (Bacillus thuringiensis):

- Limited evidence of digestive tract inflammation; minimal hazard to humans.

Treatment and Therapy:

- Recognize signs and symptoms.
- Seek immediate medical help.
- Identify the pesticide and provide label information.
- Treatment involves specific antidotes for certain pesticide classes.
- Time management is critical for effective intervention.

Therapy and Antidotes for Pesticide Poisoning

Introduction

- Antidotes are agents used to counteract the effects of poisons, specifically pesticides in this context.
- Medical antidotes should only be prescribed or administered by a physician to avoid misuse.
- Some pesticides have no known antidotes, and irreversible effects can occur after a lethal dose.

Cholinesterase and Pesticide Poisoning:

- Cholinesterase is crucial for regulating nerve impulse transmission.
- Both carbamate and organophosphate pesticides attack cholinesterase, leading to nerve impulse disruption.
- Decreased cholinesterase levels indicate pesticide overexposure.

Types of Antidotes

Cholinesterase Reactivators:

- Pralidoxime (Protopam, 2-PAM): Reactivates phosphorylated AChE, used for muscle and respiratory weakness in pesticide exposure.
- Administer within 24 hours for better results.

Anticholinergics:

- Atropine: Used for GI or pulmonary distress in pesticide poisonings.
- Continue until bronchoconstriction is controlled, and high doses may be needed in severe cases.

Diazepam (Valium):

Depresses CNS levels, caution with other depressants, monitor for respiratory depression.

Lorazepam (Ativan):

Used for status epilepticus, monitor for respiratory depression and caution in specific conditions.

Methods of Administration for Different Pesticide Groups:

Organophosphates:

- Administer atropine sulfate intravenously, maintain atropinization for 24-48 hours.
- Use diazepam for convulsions and administer cholinesterase reactivators if available.

Carbamates:

- Follow atropine therapy similar to organophosphates.
- Avoid oximes and certain medications.

Organochlorines:

- Secure airway, control convulsions with anticonvulsants, inject calcium gluconate intravenously.
- Protect vital organs with corticosteroids and dialysis if necessary.

Synthetic Pyrethroids:

- Symptomatic treatment for skin exposure.
- Perform gastric lavage for ingestion, administer activated charcoal, and control seizures with diazepam or barbiturates.

Important Considerations:

- Avoid specific medications like morphine, phenothiazines, succinylcholine, and epinephrine.
- Close observation required for patients experiencing convulsions.
- Fat-free diet with high proteins, carbohydrates, and calcium recommended in severe cases.
- Adrenaline derivatives are contraindicated.

Conclusion:

- Proper identification of pesticide poisoning, timely administration of antidotes, and careful monitoring are essential for effective treatment. Always follow physician recommendations for medication administration.